

Machine Learning Canvas v0.4 — JusticeCast

Goal [Business Understanding]: Quantify how much of bench-reading signal lives in lexical features vs semantic representations on SCOTUS oral arguments.

Decisions

[Business Understanding]

Per-Justice vote forecasts inform:

- Pre-argument prep targeting (litigators, week before)
- Amicus brief targeting (post-argument, hours after)
- Historical bench-reading benchmarks (research, continuous)

ML Task

[Modeling]

Binary stance classification per (case, Justice) row:
label = voted_petitioner $\in \{0, 1\}$
Comparative study: BoW vs pre-trained embeddings

Value Propositions

[Business Understanding]

Empirical lower bound on text-only bench-reading;
methodological evidence that pre-trained semantics outperform tuned bag-of-words by ~ 4 pp on the strict contested-cases test. Don't sell TF-IDF — sell semantic representations as the floor.

Data Sources

[Data Understanding]

SCDB Justice-Centered file (release 2025_01, Latin-1 encoded, 83,644 vote rows $\times$ 61 cols) — labels
Oyez REST API (2-step fetch: case meta $\rightarrow$ transcript audio JSON, ≤ 1 req/sec) — text
Both free, public, no auth.

Offline Evaluation

[Model Evaluation]

Test ROC AUC on canonical fold-0
5-fold CV mean $\pm$ std (StratifiedGroupKFold by case_id)
Per-Justice contested-cases AUC (the strict test)
Per-Justice lift over individual baseline

Making Predictions

[Modeling]

Per-Justice per-case forecast within hours of oral argument.
Threshold tunable per consumer use case:

- Litigators: FN-tilted (recall on petitioner side)
- Amicus: FP-tilted (precision on petitioner side)

Collecting Data

[Data Understanding]

Per-term refresh after each SCOTUS term ends.
SCDB: single annual download.
Oyez: incremental fetch of new (term, docket) pairs;
both layers cached on disk (377 MB for the 2005-2024 window).

Features

[Data Preparation]

Track 1 (BoW): TF-IDF + n-grams (200K-dim sparse)
+ 424-term custom stopword list + advocate-name regex
Track 2 (Embeddings): all-MiniLM-L6-v2 (384-dim dense)
pre-trained, no fine-tuning, no stopword list
Identical fold-0 test rows for both tracks (Non-Neg #15).

Live Evaluation & Monitoring

[Model Deployment]

Track per-term ROC AUC and per-Justice contested AUC.
Flag if AUC drops > 5 pp term-over-term — audit bench composition + case-mix shifts.
Justice composition changes require justice_id_map updates.

Building Models

[Modeling]

Annual retrain after each SCOTUS term ends (June).
Re-encode new term transcripts via compute_embeddings.py (~ 12 min CPU). Refit LogReg on full historical corpus.
Re-verify SCDB codebook semantics each major release.

Stakeholders

[Business Understanding]

Appellate litigators (pre-argument prep)
Amicus brief authors (post-argument filing decisions)
Legal-tech vendors (Lex Machina, Bloomberg Law, Westlaw Edge, SCOTUSblog) — embedded research workflow
Litigation press (same-day forecast articles)

Headline Result

[Model Evaluation]

Test ROC AUC: BoW 0.532 $\rightarrow$ Embeddings 0.569 (+3.7 pp)
Contested-cases mean per-Justice AUC: 0.532 $\rightarrow$ 0.576 (+4 pp)
Justices > 0.5 (contested): 9/15 BoW $\rightarrow$ 13/15 Embeddings
KBJackson centerpiece: 0.405 $\rightarrow$ 0.643 (+0.238) on contested.
Lightweight MiniLM beats tuned BoW on the strict test.

Each box tagged with its corresponding CRISP-DM phase. Left two columns = Predict (decisions, ML task, eval, predictions, monitoring). Right two columns = Learn (data sources, features, build cadence). Headline result corner = Model Evaluation phase.